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MIL-S-24187/7B(SH)
22 March 1991

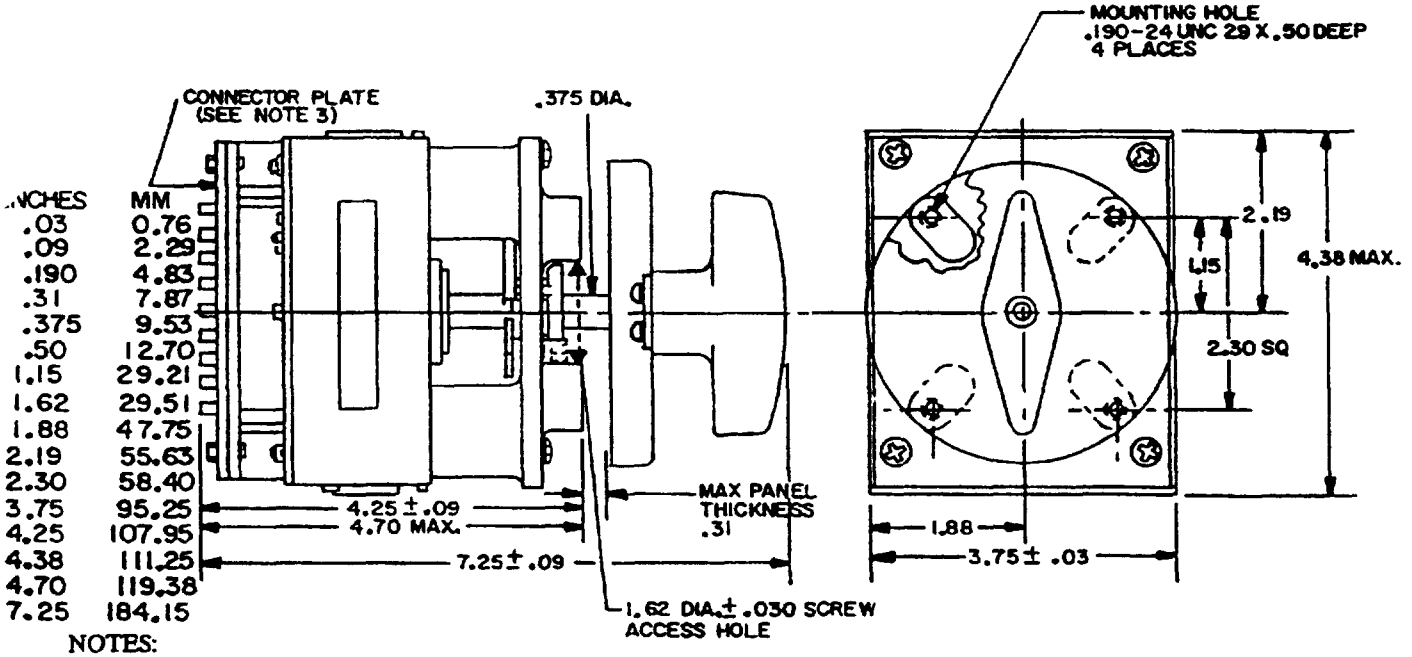
SUPERSEDING
MIL-S-24187/7A(SH)
9 September 1983

MILITARY SPECIFICATION SHEET

SWITCH ASSEMBLIES, LINEAR MOVEMENT, MANUAL OPERATIONS, STYLE KLS

This specification sheet is approved for use by the Naval Sea Systems Command, Department of the Navy, and is available for use by all departments and agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-S-24187.



NOTES:

1. Dimensions are in inches.
2. Unless otherwise specified, tolerances are plus or minus 0.02 (0.51mm) for two-place decimals and plus or minus 0 005 (0.13 mm) for three-place decimals.
3. Supplied only when specified in contract or order
4. Metric equivalents (to the nearest 0 01mm) are given for general information and are based upon 1 inch = 25 4 mm

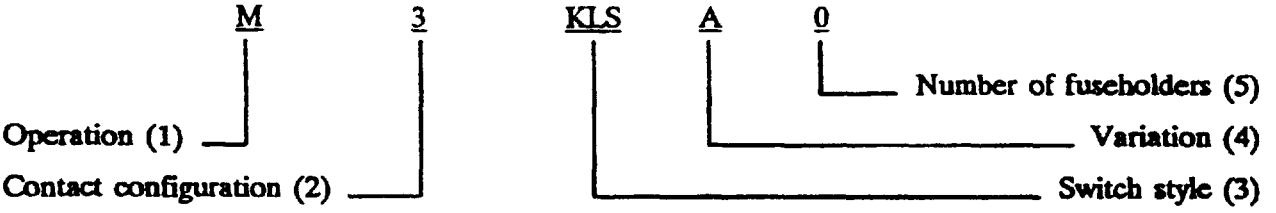
FIGURE 1 *Style KLS switch assembly*

AMSC N/A FSC 5930

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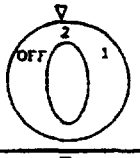
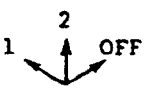
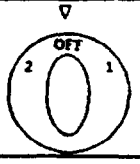
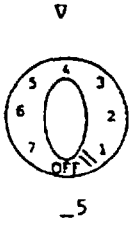
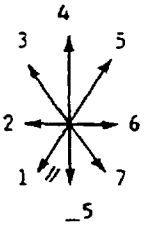
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Type designation:



(1) Switch assemblies shall be manual operation designated M.
(2) Contact configuration shall be type 2, 3, or 4 (see figure 2).
(3) Switch style shall be designated by the letters KLS.
(4) Variation in type 4 contact configuration shall be designated by the letter A.
(5) Fuseholder mounting capability shall not be provided in style KLS switches.

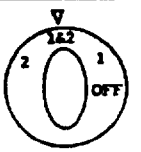
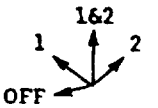
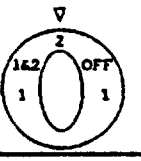
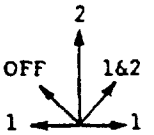
TABLE I. *Switch selection.*

Type designation				Application data					
Switch operation	Contact configuration	Switch style	Fuseholders	Switching action	Stop position[1]	Dial position[2]	Handle position[3]	Wired contacts[4]	Unwired contacts[4]
M	2	KLS	0	1-2-OFF	2&7			( )D LOAD 2 ( )C INPUT ( )B LOAD 1	( )E OFF
M	2	KLS	0	1-OFF-2	2&7			( )E LOAD 2 ( )D INPUT/OFF ( )C ( )B LOAD 1	
M	3	KLS	0	1-4-7-OFF	4			GD LOAD 7 FC LOAD 6 EB LOAD 5 DE LOAD 4 CD LOAD 3 BC LOAD 2 AB LOAD 1 XX INPUT	HE OFF

See footnotes at end of table

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TABLE I. *Switch selection – Continued.*

Type designation				Application data					
Switch operation	Contact configuration	Switch style	Fuseholders	Switching action	Stop position[1]	Dial position[2]	Handle position[3]	Wired contacts[4]	Unwired contacts[4]
M	4	KLS	0	OFF-1-1&2-2	3&7			( )E LOAD 2 ( )D INPUT ( )C LOAD 1	( )B OFF
M	4	KLSA	0	1-OFF-2 1&2-1	3&7			( )C LOAD 1 ( )D INPUT ( )E LOAD 1	( )B OFF

Example: Type designation for a manual switch assembly, type 3 is M3KLS0.

[1]Stops shall be adjustable to limit switch handle rotation to the position indicated or to any consecutive positions within the maximum rotation indicated.

[2]The flat on the handle shaft shall be horizontal and on top of the shaft when the movable contacts are positioned to close the switch circuits indicated by the V marker (see figure 2). Dial markings shown are for reference only and shall not appear on finished products unless otherwise specified in the contract or order.

[3]The handle, rotary description plate, and indexing washer assembly (see NAVSEA Drawing 803-5002990 for details) shall be adjustable to point the handle upward or to any specified position in 45-degree increments from upward. Unless otherwise specified in the contract or order, handles shall be positioned as shown.

[4]Configuration and identification of contacts shall comply with figures 2 and 3

[5]Limit of rotation.

REQUIREMENTS:

Construction: Style KLS switch parts, modules, and assemblies shall be constructed in accordance with NAVSEA Drawing 803-5002990.

Characteristics: KLS switch assemblies shall comply with characteristics in accordance with table II.

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TABLE II. *Characteristics.*

Characteristics	Values			
	Contact configuration			
	2	3	4	4—A
Number of poles	28	14	28	14
Number of positions	2 or 3	2 to 8	4	5
Break-before-make	Yes	Yes	–[1]	–[1]
Electrical rating and endurance loading, minimum:				
10 Amperes, 115 V, 60 Hz, unity p.f.	20,000 Operations	15,000 Operations	20,000 Operations	20,000 Operations
7.5 Amperes, 115 V, 60 Hz, unity p.f.	30,000 Operations	20,000 Operations	30,000 Operations	30,000 Operations
5 Amperes, 50 Vdc, resistive	30,000 Operations	20,000 Operations	30,000 Operations	30,000 Operations
2 Amperes, 115 Vdc, resistive	30,000 Operations	20,000 Operations	30,000 Operations	30,000 Operations
25 Microamperes, 1.5Vdc, resistive	30,000 Operations	20,000 Operations	30,000 Operations	30,000 Operations
Contact resistance, maximum:				
At 10 amperes	16 milliohm			
At 2.5 microamperes	50 milliohm			
Torque:				
Shaft rotation, maximum	20 pounds			
Stop strength, minimum	100 pound-inches			
Weight, maximum	3.25 pounds			
Temperature and humidity (for rotational torque and endurance tests).				
Minimum	Minus 20 °F			
Maximum	Plus 149 °F and 40 to 50 percent rh			
Shock	MIL-S-901, grade A, class I, type C			
Vibration	MIL-STD-167-1, type I, except omit variable frequency			
Temperature rise, maximum	30 °C			

[1]The 4KLS and 4KLSA shall be a combination of make-before-break (MBB) and break-before-make (BBM). Operation between the "1 and 2" position and either the "1" position or the "2" position shall not interrupt the switch circuit, "1" or "2", which is closed in both the initial and final switch positions.

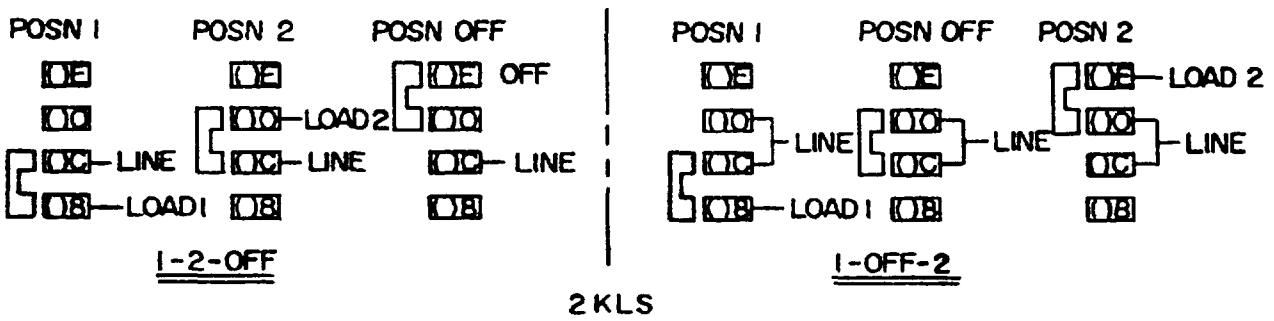
Switch assembly modules: Manual operated KLS switch assembly shall consist of the following modules as detailed on Drawing 803-5002990:

(a) Front module
(b) Switch contact module

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Switch function.

Style KSL switch assemblies shall provide the switching functions as shown on figure 2. Contact configuration 2 shall provide 28 two- or three-position poles, each consisting of four stationary contacts B, C, D, and E, and a movable contact. Contact configuration 3 shall provide 14 two- to eight-position poles, each consisting of nine stationary contacts X, A, B, C, D, E, F, G, and H, and a movable contact. Contact configuration 4 shall provide 28 four-position poles, each consisting of three stationary contacts C, D, and E, and a movable contact. The modified type 4 (4KLSA) contact configuration shall provide 14 five-position poles, each consisting of three stationary contacts B, C, and E, and a movable contact. Stops shall prevent transfer of the movable contacts into unused positions. Removable stops shall be provided which, when inserted as specified in table I, will limit movement to consecutively used positions. Stops shall be as detailed on Drawing 803-5002990 and shall be accessible from the front of the switch assembly after removal of the handle and rotary description plate.



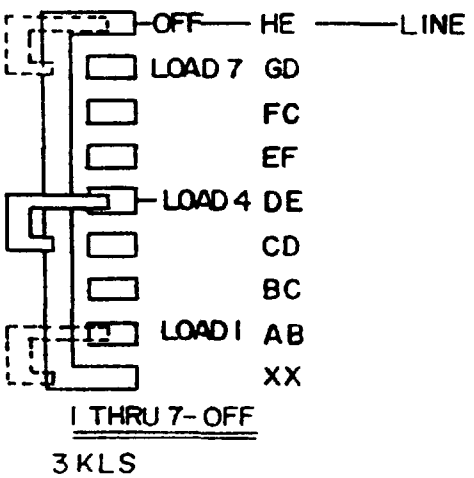


FIGURE 2. *Contact configuration*

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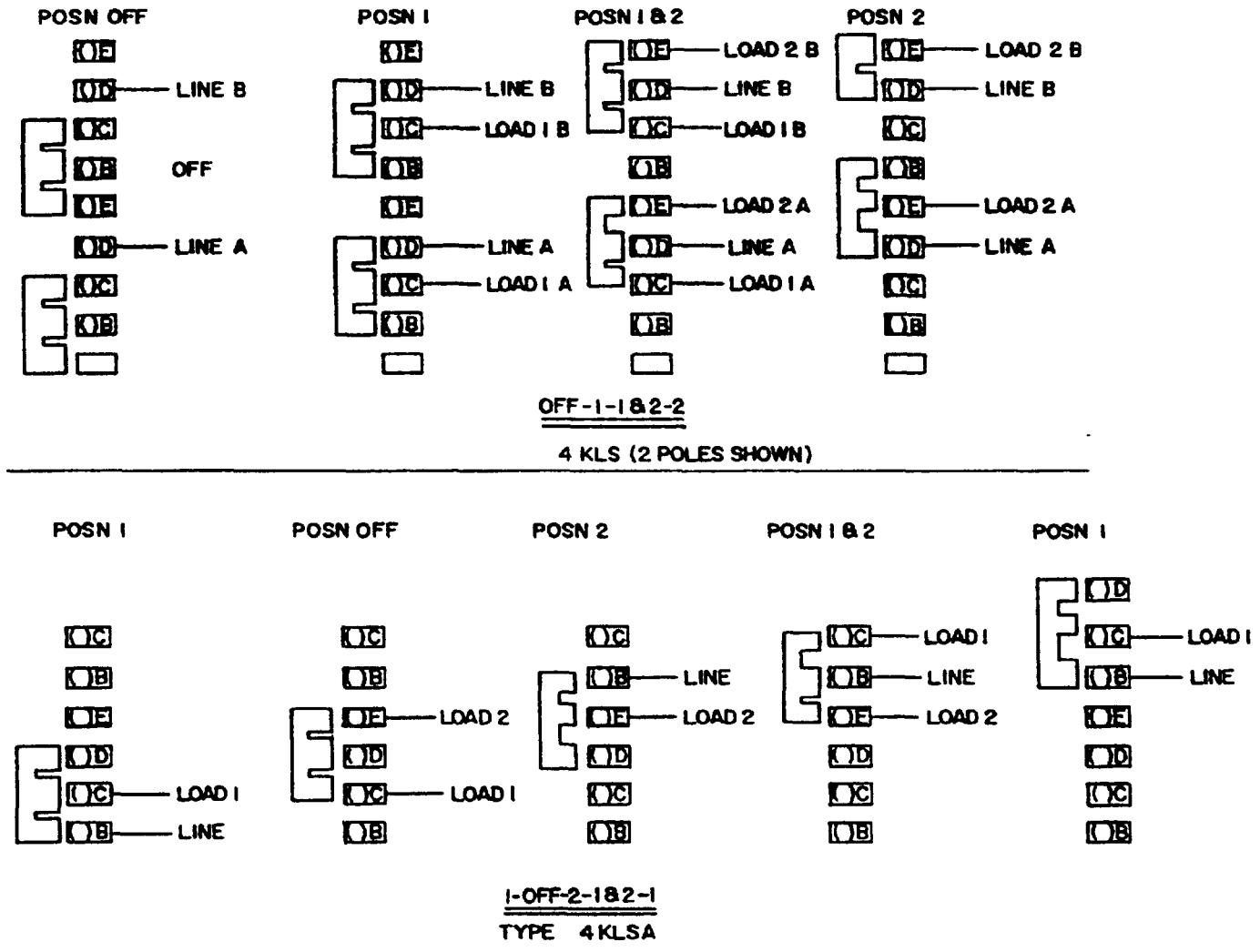
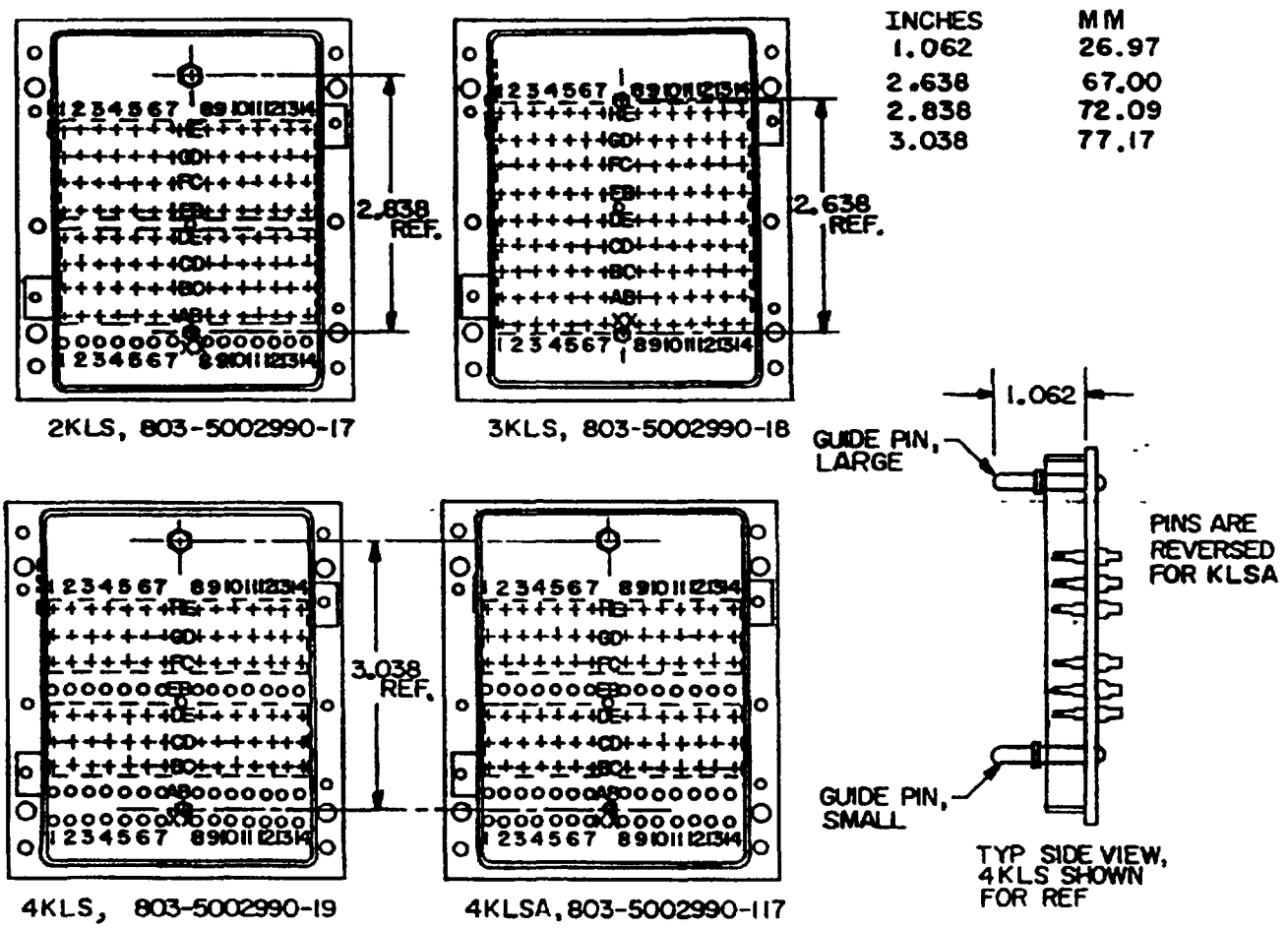


FIGURE 2. *Contact configuration – Continued.*

Contact designation: Each stationary switching contact shall be identified by two letters and a number. The first letter shall identify the pole group in which a contact is located and the second letter shall be assigned as shown on figure 2. The number shall be that of the column in which the contact is located (see figure 3) Each contact in the switch contact module connector plate shall be assigned the same identification as the contact in the switch contact module to which it connects

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NOTE: Bushings and contacts shall be installed in the stationary plates as indicated. Crosses signify bushings and contacts which are installed. Circles signify omitted bushings and contacts. Dashed rectangles, enclosing groups of poles, are shown for information only.

FIGURE 3. *Stationary plate assembly, rear view.*

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Repair parts: Repair parts shall be as specified in table III.

TABLE III. *Repair parts.*

Drawing 803-5002990 piece No.	Description	Quantities required for quantity of switches installed					
		1–3	4–12	13–30	31–60	61–120	121 or more
Manual							
1	Switch, type M2KLS0	1	1	1	1	2	3
2	Switch, type M3KLS0	1	1	1	1	2	3
3	Switch, type M4KLS0	1	1	1	1	2	3
100	Switch, type M4KLSA0	1	1	1	1	2	2
16	Detent spring	1	1	1	2	3	6
43	Stop screw	1	1	1	2	3	4

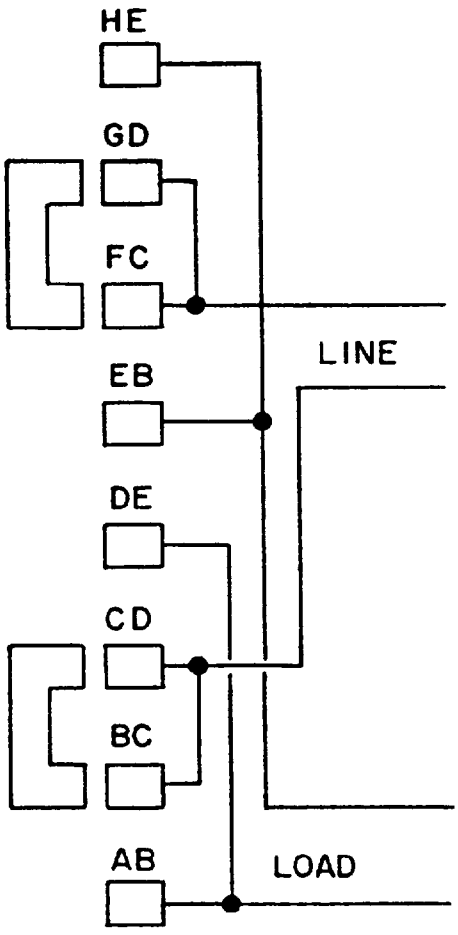
Note: Connector plates, piece numbers 73, 74, 75, and 101, are not to be supplied.

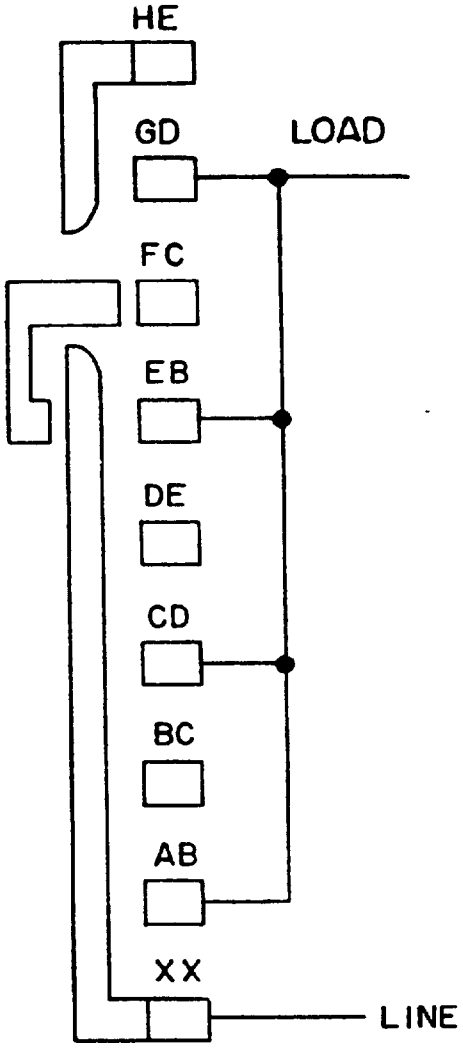
Qualification samples: Two style KLS switch assemblies of each contact configuration shall be submitted for qualification inspection.

Tests:

Loading of switches: Loading of switches for endurance and temperature rise tests for qualification and quality conformance, group C testing, shall be in accordance with figure 4 and table IV.

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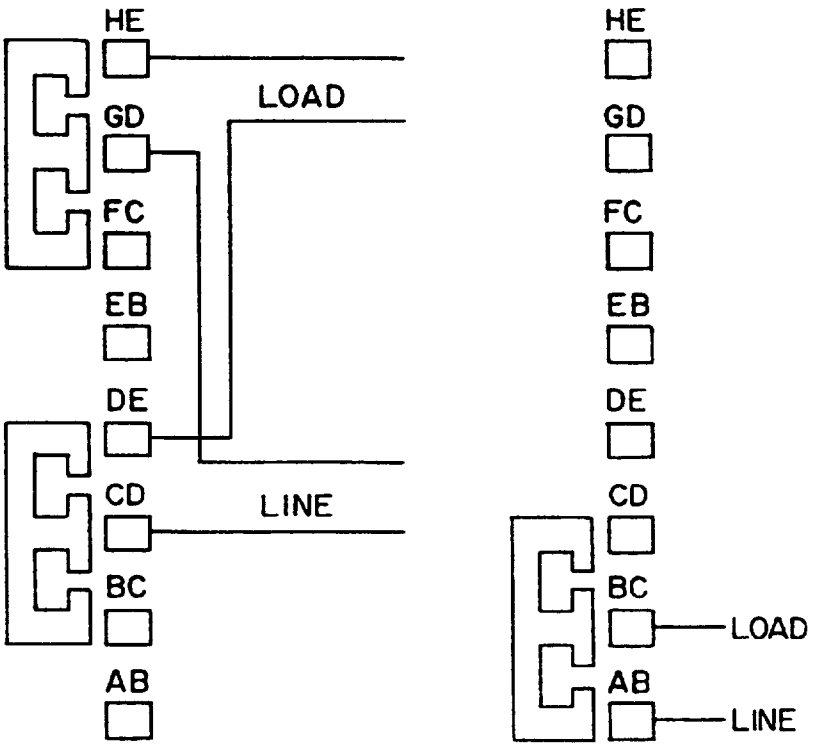


Columns	Type 2KLS Volts	Amp
1	115	10 ac
5	50	5 dc
7 and 9	115	7.5 ac
11	115	2 dc
14	1 5	25 μdc

Columns	Type 3KLS Volts	Amp
1 and 2	115	10 ac
5 and 6	50	5 dc
11 and 12	115	2 dc
9	1 5	25 μdc
13 and 14	115	7 5 ac

FIGURE 4 *Test loading*

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Type 4KLS			Type 4KLSA		
Columns	Volts	Amp	Columns	Volts	Amp
1	115	10 ac	1	115	10 ac
5	50	5 dc	5	50	5 dc
11	115	2 ac	11	115	2 dc
14	1.5	25 μdc	14	115	25 μdc
7 and 9	115	7.5 dc	7 and 9	115	7.5 ac

FIGURE 4. *Test loading – Continued*

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Temperature rise: During temperature rise test, contacts specified in table IV shall be monitored.

TABLE IV. *Temperature measurements.*

Column	Contacts			
	Type 2KLS	Type 3KLS	Type 4KLS	Type 4KLSA
1 5	EB, FC EB, FC	XX, GD XX, GD	GD, HE GD, HE	AB, BC AB, BC

Connector accessories: When specified in the contract or order, switch assemblies shall be furnished with connector accessories in accordance with figures 5, 6, 7, and 8.

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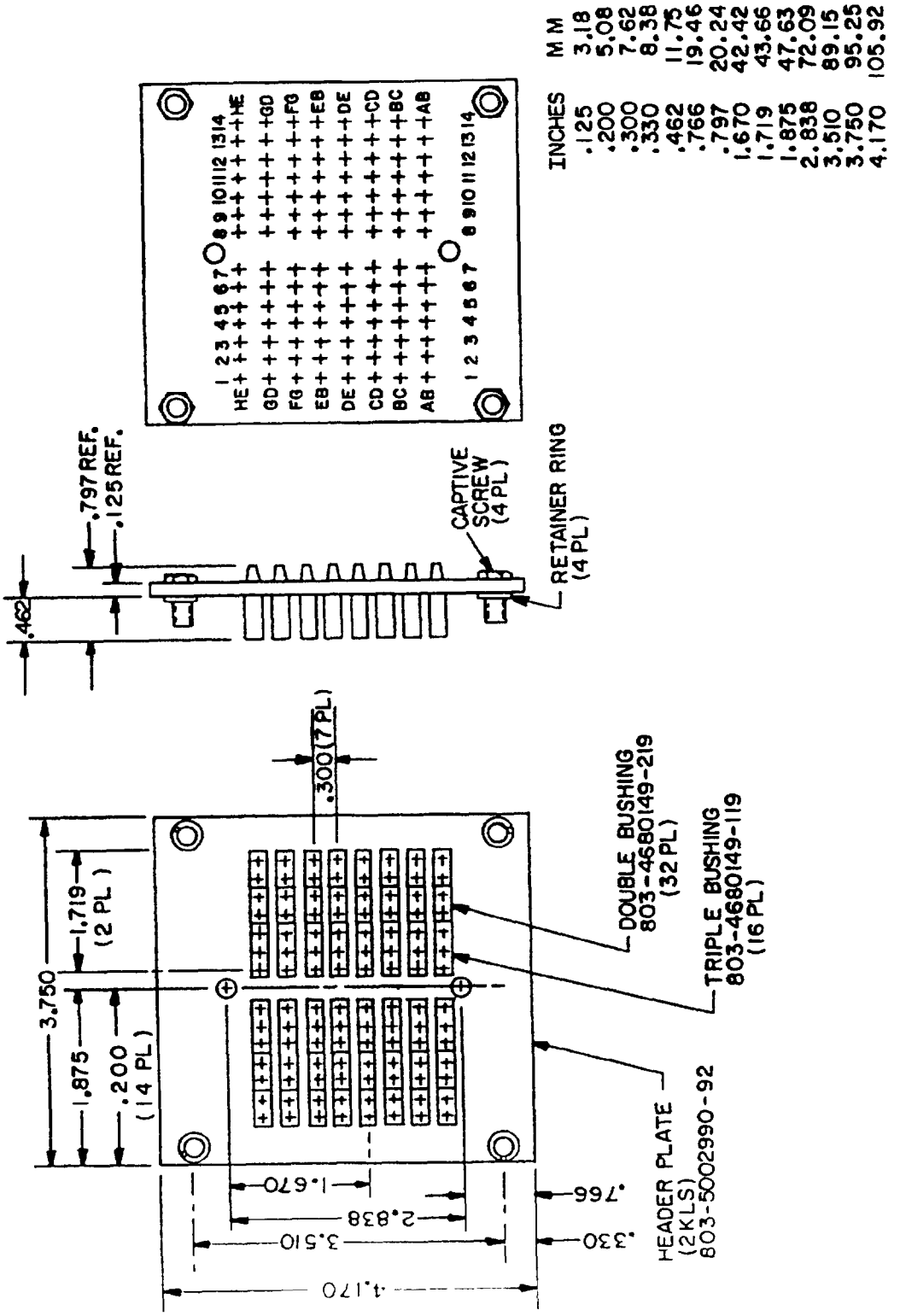


FIGURE 5. *Accessories – type 2KLS connector plate, 803-5002990-75.*

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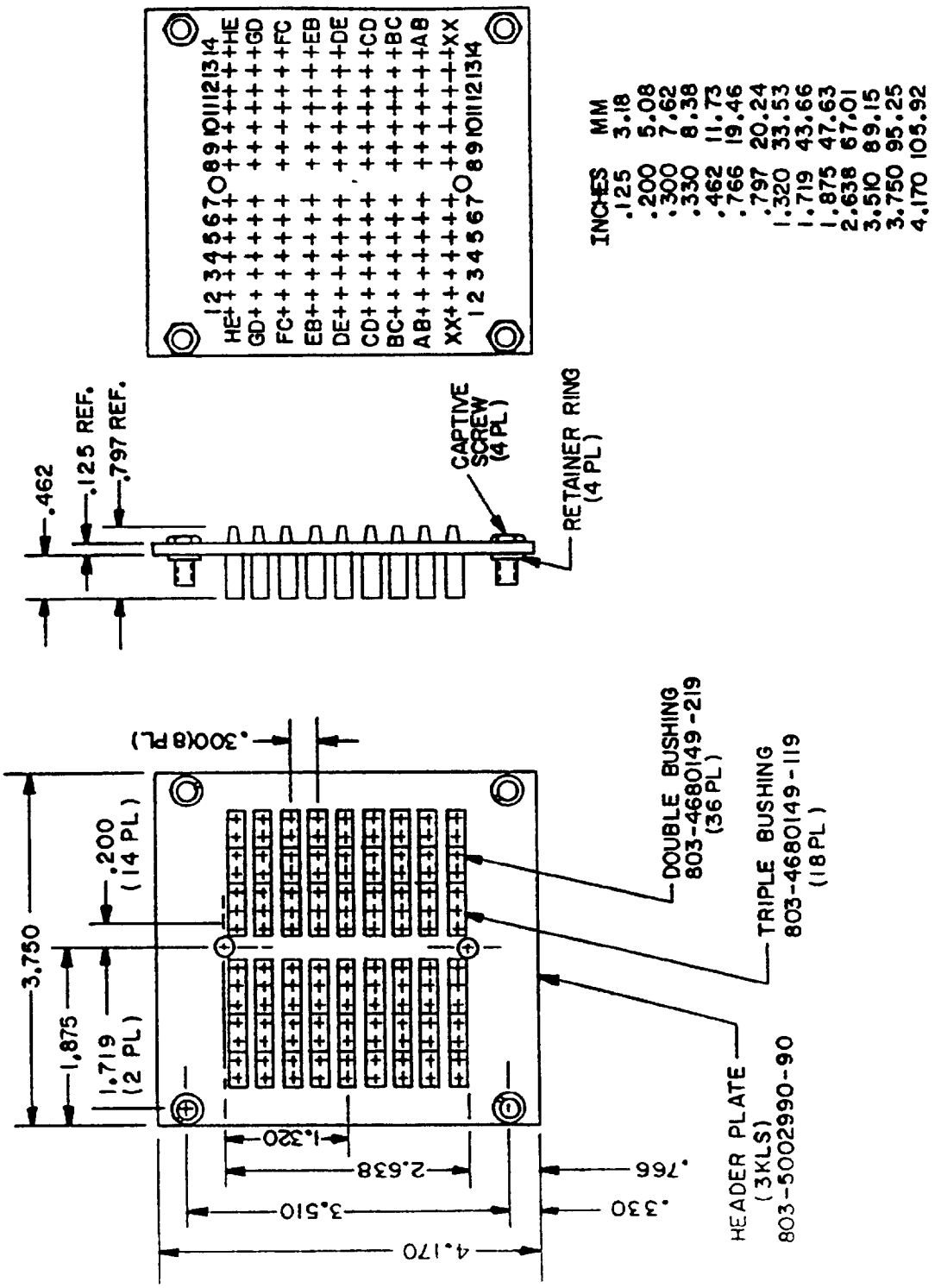


FIGURE 6. *Accessories – type 3KLS connector plate, 803-5002990-74.*

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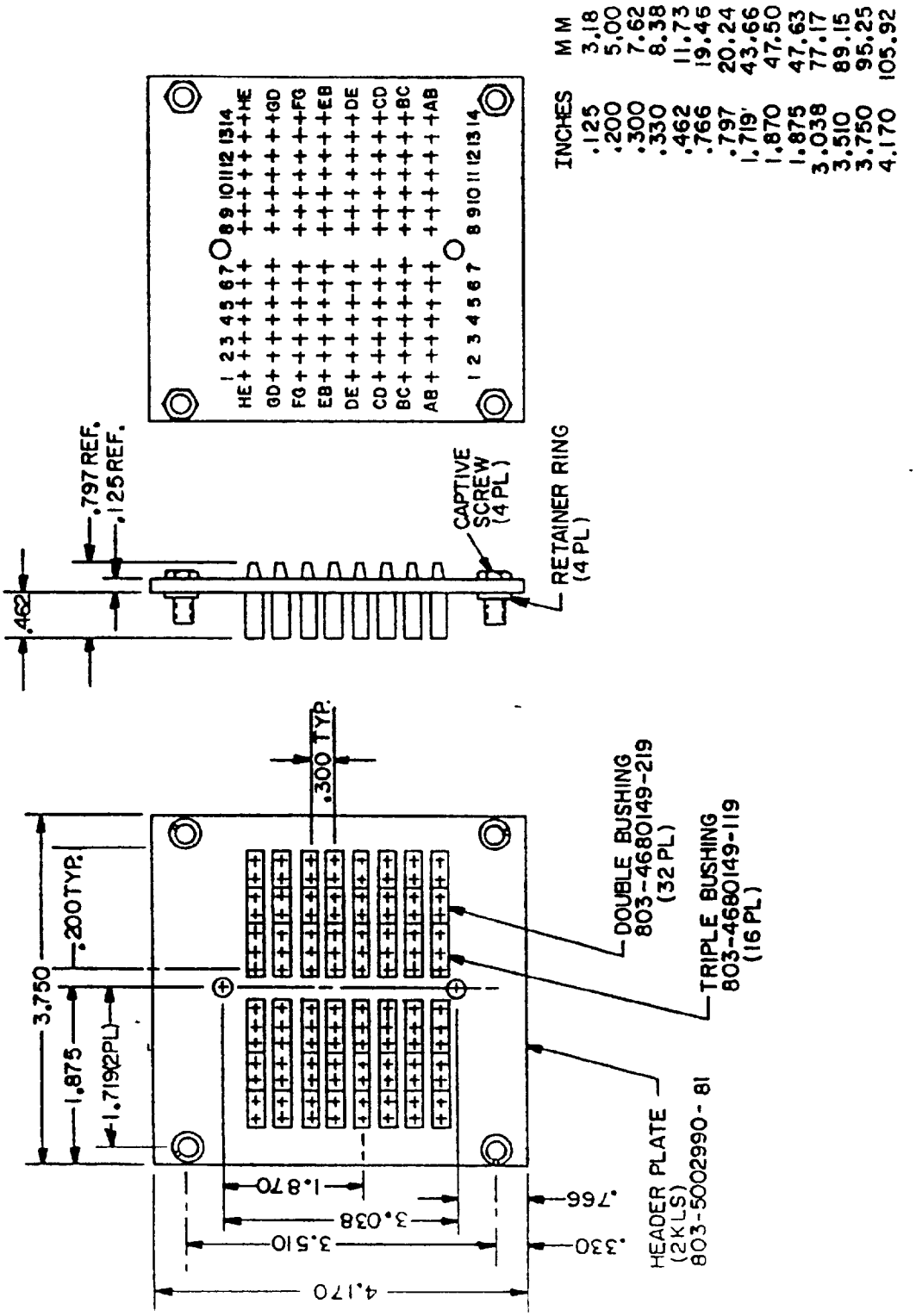


FIGURE 7. *Accessories – type 4KLS connector plate, 803-5002990-73.*

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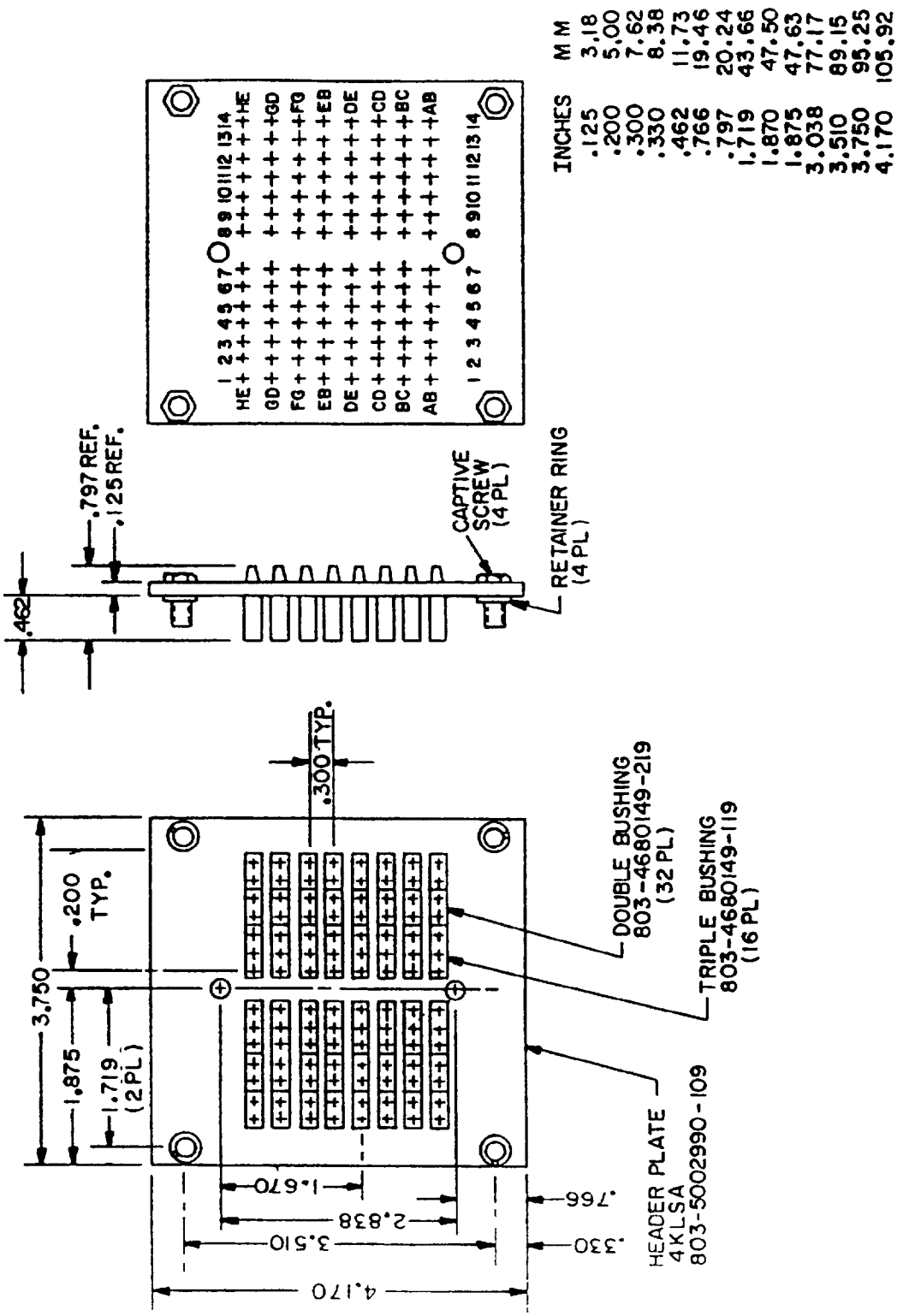


FIGURE 8. *Accessories – type 4KLSA connector plate, 803-5002990-101.*

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Preparing activity:
Navy – SH
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